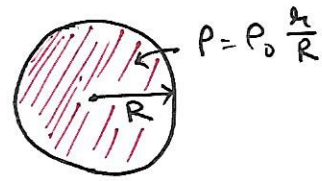
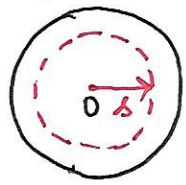
1. The solid sphere has a volume charge density $\rho = \rho_0 \frac{r}{R}$ where r is the radial distance in spherical polar co-ordinates

<u>Inside sphere:</u> $0 \leq s \leq R$



Consider a Gaussian sphere inside the sphere of radius R. Apply Gauss's law over the Gaussian surface:



$$\oiint \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} Q_{enc} \quad \text{—— (1)}$$

To find Q_{enc} within the sphere of radius 's', consider volume element $d\tau$ at $\vec{r}'$ from O

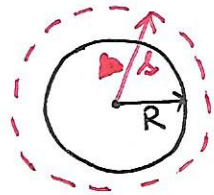
$$E_{in} \, 4\pi s^2 = \frac{1}{\epsilon_0} \int \rho(r') \, d\tau'$$ [primed variables refers to point inside 's']

$$= \frac{1}{\epsilon_0} \int_{r'=0}^{s} \rho_0 \frac{r'}{R} r'^2 \, dr' \underbrace{\int_0^{\pi} \sin\theta \, d\theta}_{2} \underbrace{\int_0^{2\pi} d\varphi}_{2\pi} \quad [d\tau' = r'^2 dr' \sin\theta \, d\theta \, d\varphi]$$

$$= \frac{1}{\epsilon_0} \int_0^s \left(\frac{\rho_0 r'}{R}\right) 4\pi r'^2 \, dr' = \frac{4\pi\rho_0}{\epsilon_0 R} \int_0^s r'^3 \, dr' = \frac{4\pi\rho_0}{\epsilon_0 R} \frac{s^4}{4}$$

$$\therefore \boxed{\vec{E}_{in} = \frac{\rho_0 s^2}{\epsilon_0 R} \hat{s}} \quad \text{—— (2)}$$ [here $\hat{s}$ is also the radial direction]

<u>Outside sphere</u> $s > R$



Consider Gaussian sphere of radius 's' such that $s > R$. Apply Gauss's law over this surface:

$$\oiint \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} Q_{enc}.$$

$$Q_{enc} = \int \rho(r') \, d\tau' = \int_{r'=0}^{R} \rho(r') \, 4\pi r'^2 \, dr'$$ [4π factor from integration of θ and φ in $d\tau'$]

Similar to above, $d\tau'$ volume element can be considered to be at $\vec{r}'$ from O inside the sphere - except limits of r' will be from 0 to R because volume charge density (source) is only confined with the sphere.

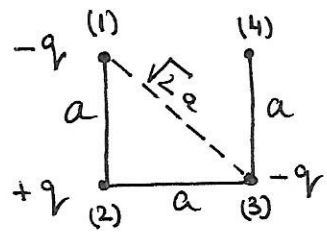
$$\therefore E_{out}\, 4\pi r^2 = \frac{1}{\epsilon_0}\int_{r'=0}^{R} \rho(\vec{r}')\, 4\pi r'^2\, dr' = \frac{\rho_0}{\epsilon_0}\frac{4\pi}{R}\int_0^R r'^3\, dr'$$

$$= \frac{4\pi\rho_0}{\epsilon_0 R}\,\frac{R^4}{4}$$

$$\therefore \boxed{\vec{E}_{out} = \frac{\rho_0 R^3}{\epsilon_0 r^2}\,\hat{r}} \quad \text{——} \quad (3)$$

Results (2) and (3) are easy to arrive at compared to the integration method adopted to the same problem in Tutorial 6. This explains the power of Gauss's law when symmetry is involved in the problem.

2



(a) Work done to bring fourth charge $+q$ and place it at position (4)

$= +q \times$ [potential at pos. 4 due to all other three charges]

$[\because W = q\,V(\vec{r})]$

$$= +q \times \left[\frac{-q}{4\pi\epsilon_0 r_{34}} + \frac{+q}{4\pi\epsilon_0 r_{24}} + \frac{-q}{4\pi\epsilon_0 r_{14}}\right]$$

Clearly $r_{14} = a$, $r_{24} = \sqrt{2}\,a$, $r_{34} = a$

$$W = \frac{+q^2}{4\pi\epsilon_0}\left[\frac{-1}{a} + \frac{1}{\sqrt{2}\,a} - \frac{1}{a}\right] = \boxed{\frac{q^2}{4\pi\epsilon_0 a}\left(-2 + \frac{1}{\sqrt{2}}\right)}$$

(b) To calculate the total work in assembling all the four charges, we would ~~and~~ consider the work ~~do~~ to bring charges one by one.

W_1 = work done to bring $-q$ at pos. 1 = 0 [as there is no potential due to other charges]

W_2 = work done to ~~from~~ bring $+q$ at 2

$= +q \times$ [pot. at 2 due to $-q$ at 1] $= -\dfrac{q^2}{4\pi\epsilon_0 a}$

$W_3 =$ work done to bring $-q$ at pos. 3

$= -q \times$ [pot. at 3 due to "$-q$ at 1" and "$+q$ at 2"]

$$= -q \left[\frac{-q}{4\pi\epsilon_0\sqrt{2}\,a} + \frac{q}{4\pi\epsilon_0 a} \right] = \frac{q^2}{4\pi\epsilon_0 a}\left(\frac{1}{\sqrt{2}} - 1\right)$$

$W_4 =$ work done to bring $+q$ at pos. 4

$= +q \times$ [pot. at 4 due to '$-q$ at 1', '$+q$ at 2', and '$-q$ at 3']

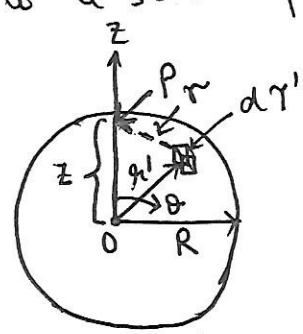
$$= +q \times \left[\frac{-q}{4\pi\epsilon_0 a} + \frac{q}{4\pi\epsilon_0\sqrt{2}\,a} + \frac{-q}{4\pi\epsilon_0 a} \right] = \frac{q^2}{4\pi\epsilon_0 a}\left[-2 + \frac{1}{\sqrt{2}}\right]$$

Total work, $W_{tot} = W_1 + W_2 + W_3 + W_4 = \dfrac{q^2}{4\pi\epsilon_0 a}\left[\cancel{-1} + \dfrac{1}{\sqrt{2}} - 1 - 2 + \dfrac{1}{\sqrt{2}}\right]$

$$\boxed{W_{tot} = \frac{2q^2}{4\pi\epsilon_0 a}\left(-2 + \frac{1}{\sqrt{2}}\right)}$$

3 For a volume charge density ρ, the electrostatic energy is given by $W = \frac{1}{2}\int_V \rho V \, d\tau$. For the derivation, see lecture notes [consider energy of a point charge distribution and then generalize to continuous charge distribution]. The region of integration is the volume enclosed by total charge.

First let's calculate the potential V due to a solid sphere of uniform charge.



Consider a volume element $d\tau$ inside the sphere at $\vec{r}'$. The potential at any point z inside the sphere is given by

$$V(z) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(r')\,d\tau'}{\mathscr{r}}$$ [$\mathscr{r}$ = relative dist. between $d\tau'$ and P]

$d\tau' = r'^2 dr' \sin\theta \, d\theta \, d\phi$ and $\mathscr{r} = (r'^2 + z^2 - 2r'z\cos\theta)^{1/2}$ (by law of cosines)

$$V(z) = \frac{1}{4\pi\epsilon_0}\int_{r'} \rho\, r'^2\, dr' \int_{\theta=0}^{\pi} \frac{\sin\theta\, d\theta}{(r'^2+z^2-2r'z\cos\theta)^{1/2}} \underbrace{\int_0^{2\pi} d\varphi}_{(2\pi)}$$

$$= \frac{2\pi\rho}{4\pi\epsilon_0}\int_{r'} r'^2 dr' \times P_\theta \quad \text{where } P_\theta = \int_0^{\pi} \frac{\sin\theta\, d\theta}{(r'^2+z^2-2r'z\cos\theta)^{1/2}}$$

Lets evaluate P_θ first.

$$P_\theta = \int_{(r'-z)^2}^{(r'+z)^2} \frac{du}{2r'z\, u^{1/2}}$$

Since limits involve $(r'-z)^2$ and $(r'+z)^2$, we have to investigate regions $r'>z$ and $r'<z$ separately.

Let $r'^2+z^2-2r'z\cos\theta = u$

$2r'z\sin\theta\, d\theta = du$

$\sin\theta\, d\theta = \frac{du}{2r'z}$

$\theta=0 \Rightarrow u=(r'-z)^2$

$\theta=\pi \Rightarrow u=(r'+z)^2$

$$P_\theta = \frac{1}{2r'z}\cdot 2\sqrt{u}\Big|_{(r'-z)^2}^{(r'+z)^2} = \frac{1}{r'z}\left[(r'+z) - |(r'-z)|\right]$$

For $\underline{r'<z}$: $|r'-z| = -(r'-z) \therefore P_\theta = \frac{1}{r'z}\left[(r'+z) + (r'-z)\right]$

$$P_\theta = \frac{2}{z} \quad (r'<z)$$

For $\underline{r'>z}$: $|r'-z| = (r'-z)$

$$P_\theta = \frac{1}{r'z}\left[r'+z - (r'-z)\right]$$

$$P_\theta = \frac{2}{r'} \quad (r'>z)$$

Because of the θ-integral depending upon $r'>z$ or $r'<z$, we will split the limits of r' in $V(z)$ from $r'=0$ to z and then from z to R

$$V(z) = \frac{2\pi\rho}{4\pi\epsilon_0}\left[\int_{r'=0}^{z} r'^2 dr' \times \frac{2}{z} + \int_{r'=z}^{R} r'^2 dr' \times \frac{2}{r'}\right]$$

(corresponds to $r'<z$ case) (corresponds to $r'>z$ case)

$$V(z) = \frac{\rho}{2\epsilon_0}\left[\frac{z^3}{3} \times \frac{2}{z} + (R^2 - z^2)\right] = \frac{\rho}{2\epsilon_0}\left(\frac{2}{3}z^2 + R^2 - z^2\right)$$

$$V(z) = \frac{\rho}{2\epsilon_0}\left(R^2 - \frac{z^2}{3}\right) \quad . \text{ But } \rho = \frac{q}{\frac{4}{3}\pi R^3} = \frac{3q}{4\pi R^3}$$

$$\boxed{V(z) = \frac{q}{8\pi\epsilon_0 R}\left(3 - \frac{z^2}{R^2}\right)}$$

Having found $V(z)$, it is easy to calculate the energy due to uniformly charged solid sphere

$$W = \frac{1}{2}\int_{\mathcal{V}} \rho V \, d\tau = \frac{1}{2}\int_{r=0}^{R} \rho \cdot \frac{q}{8\pi\epsilon_0 R}\left(3 - \frac{r^2}{R^2}\right) \cdot 4\pi r^2 dr$$

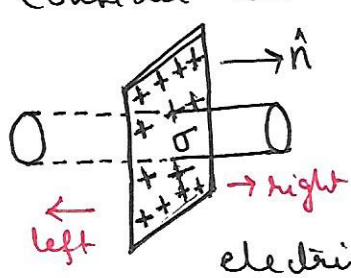
$$\left[\because V(r) = \frac{q}{8\pi\epsilon_0 R}\left(3 - \frac{r^2}{R^2}\right) \quad r < R\right]$$

$$= \frac{q\rho}{4\epsilon_0 R}\left(R^3 - \frac{R^3}{5}\right)$$

$$= \frac{q\rho}{5\epsilon_0}R^2 = \frac{qR^2}{5\epsilon_0}\,\frac{q}{\frac{4}{3}\pi R^3} = \boxed{\frac{1}{4\pi\epsilon_0}\left(\frac{3q^2}{5R}\right)}$$

4. Consider an infinitely plane sheet of charge, with charge density σ.

a

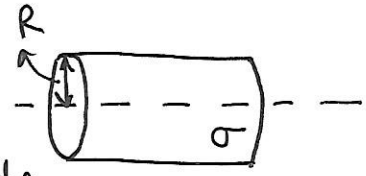


Boundary conditions for electric field says that at a point just inside and just outside the surface which carries a charge σ, the electric field is discontinuous by σ/ϵ_0.

Let us consider the direction to the right of the infinite plane sheet as positive $\hat{n}$ direction.

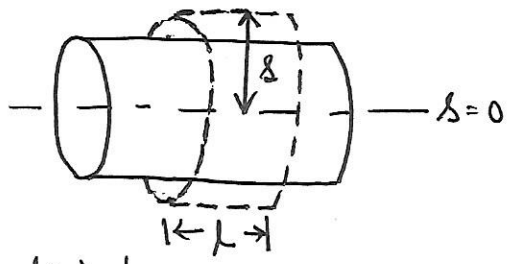
Consider a cylindrical Gaussian surface on the right and left of the interface [$\vec{E}$ is along outward area normal of the circular end of the cylinder on both sides]. $\vec{E}_{right} = \frac{\sigma}{2\epsilon_0}\hat{n}$ and $\vec{E}_{left} = \frac{\sigma}{2\epsilon_0}(-\hat{n})$. $\vec{E}_{right} - \vec{E}_{left} = \left[\frac{\sigma}{2\epsilon_0} - \left(-\frac{\sigma}{2\epsilon_0}\right)\right]\hat{n} = (\sigma/\epsilon_0)(\hat{n})$

b For a hollow cylindrical tube (radius R) carrying a uniform surface charge σ.



$E_{inside} = 0$ from Gauss's law

To calculate the field outside cylinder, consider a Gaussian cylinder of radius s and length 'l':



$$\oint\!\!\oint \vec{E}_0 \cdot \vec{da} = \frac{1}{\epsilon_0} Q_{enc}$$

$$E_0 \cdot 2\pi s l = \frac{1}{\epsilon_0} \cdot 2\pi R l \times \sigma$$

$$\boxed{\vec{E}_0 = \frac{\sigma R}{\epsilon_0 s}\hat{s}}$$ field outside cylindrical tube.

Thus at the surface $E_{sur} = \left.\frac{\sigma R}{\epsilon_0 s}\right|_{s=R} = \frac{\sigma R}{\epsilon_0 R} = \frac{\sigma}{\epsilon_0}$

$\therefore$ $E_{sur}(s \to 0^+) - E_{sur.}(s \to 0^-)$ [difference of electric fields just inside and just outside the surface]

$$= \frac{\sigma}{\epsilon_0} - 0 = \frac{\sigma}{\epsilon_0}$$

Thus the boundary conditions for electric field are always satisfied.